PREM CHANDU MEDEPALLI

Forward Deployed AI Engineer | GenAI & Agentic Systems | AI-Driven Development

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PROFESSIONAL SUMMARY

Forward Deployed AI Engineer with 2+ years at SAGE IT (since Aug 2024) who works directly with business stakeholders and clients to turn ambiguous operational problems into production GenAI systems — from discovery and rapid prototype to deployed, adopted product. Builds AI-natively: uses agentic coding tools (Claude Code, Cursor, Kiro) with spec-driven, test-backed workflows to ship full-stack platforms in days to weeks instead of months. Shipped six enterprise platforms end-to-end — AI recruitment & interview suite, vendor management portal, AI governance platform (~37% LLM cost reduction), multi-tenant AI agent SaaS, conversation intelligence for an Australian university client, and an email-to-call AI voice agent — across Python/FastAPI, .NET, React, Angular, Azure OpenAI, Azure AI Foundry, Claude and Twilio. Deep domain understanding of staffing and recruitment operations. Best Employee of the Year and 3 Spot Awards within the first year.

CORE COMPETENCIES

Forward Deployed Engineering • Customer Discovery & Solutioning • Rapid Prototyping to Production • AI-Driven / Agentic Development • LLM Applications & AI Agents • RAG & MCP • Voice AI • Real-Time Systems • Full-Stack Delivery • LLM Evaluation & Cost Governance

TECHNICAL SKILLS

- **GenAI / AI-ML:** LangChain, LangGraph, LangSmith, RAG, MCP, AI Agents, Azure AI Foundry Agents, Prompt Engineering, Embeddings, pgvector, Qdrant, Sentence Transformers, n8n
- **Computer Vision & Real-Time AI:** MediaPipe, TensorFlow.js (COCO-SSD), WebRTC, SignalR, WebSockets, Azure AI Speech (STT / Neural TTS), Whisper, IRT adaptive testing
- **Voice AI & Telephony:** Twilio Programmable Voice, ConversationRelay, Deepgram STT, ElevenLabs TTS, barge-in / interruption handling, answering-machine detection
- **LLM Platforms:** Azure OpenAI (GPT-4.1, GPT-4, Whisper, Embeddings), OpenAI API, Anthropic Claude API, Groq
- **Languages:** Python, Java, C#, TypeScript, JavaScript, SQL
- **Backend:** FastAPI, .NET 7 (Clean Architecture, MediatR, EF Core), Spring Boot, async SQLAlchemy, Alembic, Celery, Hangfire, REST APIs, JWT / RBAC, Redis
- **Frontend:** React 18/19, Angular 15, TypeScript, Vite, Tailwind CSS, TanStack Query, Recharts, Bootstrap, VS Code Extension API
- **Databases:** PostgreSQL (pgvector), MySQL, MongoDB
- **Cloud & DevOps:** AWS, Azure (Container Apps, Blob Storage, Service Bus, App Insights), Azure DevOps CI/CD, Docker, Git, Vercel
- **AI-Driven Development:** Claude Code, Cursor, Kiro, Trae, GitHub Copilot, MPCH Agent Platform; spec-driven development, agentic workflows, AI-assisted code review & testing

PROFESSIONAL EXPERIENCE

SAGE IT India Private Limited, Hyderabad, India **Aug 2024 – Present**

Associate Software Engineer (Apr 2025 – Present) • Trainee Java Full Stack Developer (Aug 2024 – Mar 2025)

- Operate as a forward deployed engineer: embed with recruiters, sales and client teams to map workflows, identify high-value AI use cases, and own delivery from requirements through deployment, adoption and iteration.
- Shipped six enterprise AI platforms and multiple client POCs end-to-end, using an AI-driven development workflow (Claude Code, Cursor, spec-driven plans) to deliver production-ready full-stack features significantly faster than traditional cycles.
- Promoted from Trainee Java Full Stack Developer to Associate Software Engineer within 7 months of joining (26 Aug 2024).
- Evaluated 8+ LLM variants for client use cases and presented Boomi AI Standards (grounding, prompt-engineering best practices, RAG architecture) to engineering teams.

PROJECTS

LeadLine AI — *Email-to-Call AI Voice Agent Platform*

Tech: FastAPI (async), SQLAlchemy 2 + asyncpg, PostgreSQL, Alembic, Twilio Voice + ConversationRelay (Deepgram STT, ElevenLabs TTS), Azure OpenAI GPT-4.1 / Groq, React + Vite + TypeScript, Tailwind CSS, WebSockets

- Built an end-to-end lead-to-call pipeline: an IMAP poller ingests lead emails, an LLM parser (with regex / phone-number fallback) extracts contact details, creates the lead, and a concurrency-capped call queue automatically dials it via Twilio.

- Implemented a real-time, human-sounding voice agent over Twilio ConversationRelay WebSockets with streamed LLM responses and barge-in — the agent stops talking, listens and responds when the caller interrupts — plus a turn-based fallback mode for trial accounts.
- Added inbound call handling that converts every caller into a lead and writes the collected details (name, company, requirement, timeline, budget, callback time) back to the lead after the call.
- Delivered post-call AI analysis (summary, sentiment, interest, radar scores), dual-channel recordings, a live transcript dashboard over WebSockets, JWT/RBAC, a full audit trail, column-selectable CSV/XLSX exports, and an LLM provider switchable from the UI.

AI Recruit — *AI Recruitment & Interview Platform*

Tech: FastAPI, React 19, PostgreSQL + pgvector, Celery, Azure OpenAI, Azure AI Foundry, Azure AI Speech, MediaPipe, TensorFlow.js, WebRTC

AI-Proctored Interview

- Built in-browser AI proctoring: MediaPipe face checks (no face, multiple faces, looking away), TensorFlow.js phone/book detection, blocked-camera and multiple-voice alerts.
- Added ID verification, fullscreen and tab-switch lockdown with 3-strike auto-submit, GPS/IP geolocation, and chunked screen recording to Azure Blob.
- Generated question banks from the JD by skill and difficulty (text, coding, whiteboard), graded with rubric-based LLM scoring, multimodal whiteboard grading and PDF scorecards.

Live Interview (Human + AI Assist)

- Built real-time video interviews on WebRTC with a FastAPI WebSocket signaling server and CoTURN, including screen share, chat, recording and live per-speaker transcription.
- Added an AI interviewer assistant with adaptive questioning (2PL IRT, Bayesian ability estimation), real-time answer scoring, and AI transcript annotations (strength, concern, red flag).

Fully AI-Driven Live Interview

- Built an autonomous voice AI interviewer that runs the full interview using an LLM state machine with rolling conversation memory.
- Used Azure real-time STT and Neural HD TTS, with echo suppression while the AI speaks and live recruiter controls to start, advance or end the interview.

Matching & Platform

- Built AI resume parsing and Claude-based candidate matching (pgvector skill similarity, eligibility gates, LLM-judged depth), deployed to Azure Container Apps via Azure DevOps.

AI Code Compiler — *Secure Coding Assessment Engine*

Tech: Python, FastAPI, Azure OpenAI, React / Angular code editor, C#

- Built a sandboxed code execution service for candidate coding rounds: isolated subprocess, CPU/memory/process limits, an unprivileged user, a 6s timeout, and fail-closed behavior in production.
- Designed hybrid evaluation: real stdin/stdout test-case judging for Python and a strict LLM-based judge for JavaScript, Java, C++ and Go; real results override LLM grades.
- Recorded every run attempt so the LLM can score the candidate's problem-solving process (debugging steps, complexity, edge cases, readability).
- Implemented MOSS-style code-plagiarism detection (tokenized k-shingles, Jaccard similarity) across candidates, ported from Python to C# for VMP.

VMP — Vendor Management Portal — *AI-Enabled Staffing & Vendor Submission Platform*

Tech: .NET 7, Angular 15, PostgreSQL, SignalR, Hangfire, FastAPI AI microservice, Azure OpenAI GPT-4.1, Azure AI Speech, Azure Blob

Vendor Portal

- Built vendor and requirement workflows for client job postings, vendor submissions, tier rules, status rounds, and Ceipal ATS integration with real-time SignalR notifications.

AI Live Interviewer

- Built an AI video interviewer on .NET 7, SignalR signaling, WebRTC/coturn and Azure Speech STT/TTS, using an LLM phase state machine with echo detection and recruiter override.
- Implemented a 2PL Item Response Theory engine in C# (Newton-Raphson MAP updates) to generate adaptive questions matched to the candidate's estimated ability.

AI Proctored Interview

- Delivered Angular browser proctoring (MediaPipe, TensorFlow.js COCO-SSD) with ID capture and IP geolocation, plus Hangfire-scheduled LLM scoring, Whisper transcription and QuestPDF scorecards.

AI Microservice

- Built a stateless FastAPI AI service (Azure OpenAI / AI Foundry agents) for weighted JD–resume matching, JD and resume analysis, and DOCX resume template formatting.

Recruitment Process Automation & AI Solutions — *Staffing & Recruitment Workflow Optimization*

- Applied hands-on business understanding of staffing and recruitment operations to identify manual bottlenecks across sourcing, screening, and client-requirement fulfillment, and designed a portfolio of automation and AI-driven solutions to streamline the end-to-end recruitment process.
- Delivered multiple recruiter-facing tools spanning workflow automation (n8n) and AI/LLM-based decisioning, reducing manual effort in requirement intake, candidate shortlisting, and resume-to-JD matching.

AI Resume-to-Template Converter & Playground Editor — *Automated Resume Formatting Tool*

- Built an AI pipeline that parses an uploaded resume and auto-generates a formatted template with ~90% accuracy, sharply cutting the manual formatting effort recruiters previously handled for every resume.
- Designed a Word-like, in-browser playground editor so users can make the remaining ~10% of corrections directly on the generated template, after which it is finalized as a reusable, ready-to-apply 100% template.
- Enabled one-click template-based conversion — once a template is finalized, any newly uploaded resume can be reformatted into the selected template automatically, standardizing resume presentation at scale.

AI Voice Calling Agent for Candidate Screening — *End-to-End Outbound Recruiter Calling Agent*

- Architected an end-to-end AI calling agent that takes a job description (JD) and a candidate's mobile number as input, autonomously places the call, and converses with the candidate using natural, human-like neural text-to-speech (TTS).
- Configured outcome-driven conversation flows so the agent tailors its screening questions and call logic to the JD requirements and predefined evaluation criteria, reducing manual recruiter screening calls.

MPCH Agent Platform — *Enterprise AI Access & Governance Platform*

- Designed and built an enterprise AI governance platform from scratch — FastAPI backend, React admin dashboard, and a custom VS Code extension — giving teams centralized access to AI models with quotas, per-user model assignment, and token-level audit trails.
- Adopted by the internal engineering team, replacing individual AI subscriptions and **cutting monthly LLM spend by ~37%** through centralized usage control and per-user cost visibility.
- Built a VS Code extension (TypeScript) with an agentic coding assistant — supervisor loop, task classification, multi-turn chat, live agent timeline, and one-click revert — packaged as a .vsix for enterprise deployment.
- Implemented a multi-provider AI proxy (Anthropic Claude + Azure OpenAI) with JWT auth, role-based permissions, rate limiting, and cost-analytics dashboards with per-user and per-organization breakdowns.

MPCH Agent Builder — *Multi-Tenant SaaS Platform for AI Agents, RAG & MCP*

- Architected a multi-tenant SaaS backend (FastAPI, async SQLAlchemy) with strict tenant isolation across 11 domain modules and 51 REST endpoints, following a layered router–service–repository pattern.
- Built the RAG module end-to-end: Qdrant vector search, sentence-transformer embeddings, hybrid BM25 retrieval, and document ingestion for PDF, DOCX, PPTX, and XLSX.
- Developed an MCP server-management system with automatic OpenAPI-spec discovery, converting arbitrary REST APIs into callable agent tools with auto-extracted parameters and schemas.
- Shipped Stripe usage-based billing with tiered plans and per-org limits, a full RBAC system, and a 27-page React/TypeScript frontend including a separate super-admin portal for cross-tenant management.

Sentilytics — *AI-Powered Sentiment & Conversation Intelligence Platform*

- Architected and delivered a full-stack sentiment analysis platform for an **Australian university client**, processing audio and text feedback through Azure OpenAI GPT-4 and Whisper into cross-modal sentiment reports, intent classification, and confidence-scored recommendations.
- Designed a custom audio signal-processing pipeline (librosa: pitch, volume, speech rate, silence detection) with k-means speaker diarisation detecting 2–5 speakers per conversation, enabling GPT-4-based sarcasm and conflict detection.
- Built multi-channel ingestion — audio uploads (5 formats), bulk CSV import, anonymous public feedback, and social media collection (Reddit, Bluesky, Facebook Graph API) with keyword matching across 16 institutions.
- Engineered granular RBAC (25+ permissions across 5 roles) with permission-gated sensitive-data masking, plus per-request AI cost attribution surfaced in an admin analytics dashboard.

Additional AI Solutions — *Internal Tools & Client POCs*

- **AI Interview & Call Agent System:** fully AI-driven interview platform with dynamic question generation, audio transcription, and automated candidate evaluation; **replaced the third-party TechScreen tool, saving ~$1.2K/year**.
- **Test Case Generator:** generates structured test cases directly from user stories with Excel export, **reducing tester effort by 80%**.
- **Resume–JD Recommendation Engine:** multi-service matching system producing bidirectional resume-to-JD recommendations with AI-based compatibility scoring.
- **Mule-to-Boomi Code Transformer:** AI-driven code conversion service that migrates Mule integrations to Boomi while preserving structural integrity.

ACADEMIC PROJECTS

Customer Churn Prediction using Machine Learning | GitHub

- Predicts whether a customer will continue or discontinue a service, trained on a telecommunications dataset.

Rock or Mine Prediction using Machine Learning

- Classifies sonar returns as rock or mine from a frequency-based dataset.

INTERNSHIPS

AWS Cloud Virtual Internship, AICTE (Virtual) | Certificate — Oct – Dec 2021

- Hands-on experience with core AWS services, tools and cloud best practices alongside industry professionals.

AI-ML Virtual Internship, AICTE (Virtual) | Certificate — Mar – May 2022

- Studied AI/ML foundations and practical applications through guided, real-world machine-learning exercises.

AWARDS & RECOGNITION

- **Best Employee of the Year** – SAGE IT, awarded within the first year of career for exceptional delivery and impact across AI/ML initiatives.
- **Spot Award ×3** – for outstanding contributions across multiple AI and development initiatives.

CERTIFICATIONS

- AWS Certified Developer – Associate • AWS Certified Cloud Practitioner • Microsoft Azure Fundamentals
- Red Hat Certified Enterprise Application Developer • Oracle Certified Foundation Associate

EDUCATION

B.Tech in Computer Science — **2020 – 2024**
KL University, Vijayawada | CGPA: 9.01

Intermediate (12th Grade) — **2018 – 2020**
Narayana Junior College, Gudivada, Andhra Pradesh | CGPA: 9.25

10th Grade (Secondary Education) — **2017 – 2018**
Montessori (EM) High School, Gudivada, Andhra Pradesh | CGPA: 9.2

PUBLICATIONS

- Crop Disease Identification using Convolutional Neural Networks – IEEE ICICT, June 2023. | Paper | DOI
- Assessing Crop Disease and Recommending Pesticide Application – IJMRSET, April 2024. | Paper | Journal Issue

LEADERSHIP

- Director of Technology, Mayavi Technology Club, KL University (2022–2023) – led AR/VR, UI/UX, and gaming initiatives; organized hands-on workshops for students. | Certificate